

Deep Neural Networks - Mid-Semester 1 Mock 01

Contents

DNN Midsem 1 Mock 01 - Question Paper	3
Q1 - 4 marks	3
Q2 - 4 marks	3
Q3 - 4 marks	3
Q4 - 4 marks	3
Q5 - 4 marks	3
Q6 - 4 marks	4
Q7 - 3 marks	4
Q8 - 3 marks	4

DNN Midsem 1 Mock 01 - Question Paper

Time: 90 minutes | Marks: 30 | Closed book

For every calculation-bearing numerical or applied response, show the formula, substitutions, intermediate steps, units, shapes, conditions, and interpretation. For a qualitative applied item, you may give reasoned N/A for formula/substitution/calculation, but must state assumptions or conditions and interpret your answer.

Q1 - 4 marks

Sessions: 1 | Topics: perceptron, threshold, convergence Point difficulty mix: Easy 2, Standard 2, Stretch 0 Provenance: Authored BITS-style | Time guidance: 10 minutes

A binary perceptron uses $z = w^T x + b$ and the course threshold convention. (a) State the threshold rule and all conditions required for the finite-mistake convergence guarantee. (2) (b) Starting with $w = (0.5, -0.2)$, $b = -0.1$, learning rate 0.2, and a mistaken example $x = (2, 3)$, $y = 1$, $\hat{y} = 0$, compute the updated parameters. (2)

Q2 - 4 marks

Sessions: 2 | Topics: affine composition, shapes, parameters Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 10 minutes

(a) Compare width and depth, and explain algebraically what happens when two dense hidden layers have no nonlinear activation. (2) (b) For a $5 \rightarrow 4 \rightarrow 2$ dense network, state every weight/bias shape and calculate the total trainable parameters. State the layer-count convention used. (2)

Q3 - 4 marks

Sessions: 3 | Topics: forward pass, chain rule, stable softmax Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

(a) For $x = 2$, $z = 2x - 1$, $h = \text{ReLU}(z)$, $L = (h - 6)^2$, compute all forward values and dL/dx by labelled local derivatives. (2) (b) For logits $(3, 1, 0)$, compute stable softmax probabilities to three decimals and cross-entropy when the second listed class is correct; briefly interpret that loss. (2)

Q4 - 4 marks

Sessions: 4 | Topics: optimization families, Adam Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

(a) State what distinguishes a saddle from a plateau. (1) (b) Compare Nesterov momentum with RMSprop by gradient location/history and scaling. (1) (c) At Adam step $t = 1$, use $g_1 = -2$, zero moments, $\beta_1 = 0.9$, $\beta_2 = 0.99$, $\eta = 0.01$, and negligible epsilon. Calculate $m_1, v_1, \hat{m}_1, \hat{v}_1$ and the parameter change; state why correction is present. (2)

Q5 - 4 marks

Sessions: 5 | Topics: split roles, overfitting, regularization Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

Losses by checkpoint are training $(0.70, 0.41, 0.24, 0.12)$ and validation $(0.73, 0.46, 0.39, 0.55)$. (a) Contrast validation with test use. (1) (b) Diagnose the trend, choose a checkpoint, and find the checkpoint-4 train-validation gap. (2) (c) Give a response and limit; BN is no universal shift cure. (1)

Q6 - 4 marks

Sessions: 6 | Topics: dilated convolution, shape, parameters Point difficulty mix: Easy 1, Standard 2, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

An input is $25 \times 19 \times 4$. A cross-correlation layer uses 7 kernels of size 3×3 , padding (2,1), stride (2,2), dilation (2,1), and one bias per output channel. (a) Compute both effective kernel sizes, output shape, and parameter count. (2) (b) Explain how dilation and stride separately caused the resulting spatial size and why image dimensions do not enter the parameter count. (2)

Q7 - 3 marks

Sessions: 7 | Topics: residual networks, transfer learning Point difficulty mix: Easy 1, Standard 2, Stretch 0 Provenance: Authored BITS-style | Time guidance: 10 minutes

You have 1,200 labelled thermal images and a model pretrained on ordinary colour photographs. Design a staged freeze/fine-tune experiment, identify one domain-mismatch risk and one overfitting check, and explain how a residual shortcut's shapes must be handled if a block halves spatial size and doubles channels.

Q8 - 3 marks

Sessions: 8 | Topics: integrated diagnosis Point difficulty mix: Easy 1, Standard 1, Stretch 1 Provenance: Authored BITS-style | Time guidance: 12 minutes

A deep sigmoid network has high training and validation loss, early-layer gradients near zero, and no improvement after adding dropout. (a) State two conditional properties that distinguish representation, optimization, and generalization claims in this case. (2) (b) Give one evidence-based diagnosis and one bounded next experiment. (1)

End of paper - 30 marks